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Larson et al.

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(54) **INTERIOR REARVIEW MIRROR ASSEMBLY WITH OVERMOLDED METAL MOUNT AND ANTI-CAMOUT TAB**

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Related U.S. Application Data

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(51) **Int. Cl.**
G02B 7/182 (2021.01)
B60R 1/04 (2006.01)

(52) **U.S. Cl.**
CPC ***G02B 7/182*** (2013.01); ***B60R 1/04*** (2013.01)

(58) **Field of Classification Search**
CPC G02B 7/182; B60R 1/04
USPC 359/872
See application file for complete search history.

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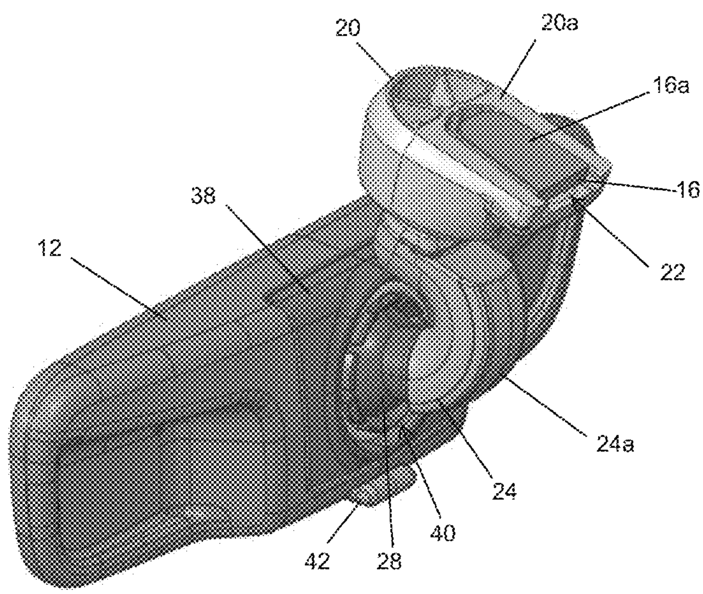
(57) **ABSTRACT**

A vehicular interior rearview mirror assembly includes a mirror head and a mounting assembly having a mirror mount and a mounting arm, which is configured to attach at an interior portion of a vehicle. The mirror head includes a mirror reflective element. The mounting arm is attached to the mirror head and the mounting arm is pivotally attached at the mirror mount via a first pivot joint. The mirror mount includes an anti-camout element protruding from an interior surface of a socket element of the first pivot joint and at least partially received in a recess of a ball member of the first pivot joint to limit pivotal movement of the ball member when the ball member engages the anti-camout element. The mirror mount comprises a metallic support element forming at least a portion of the anti-camout element.

14 Claims, 13 Drawing Sheets



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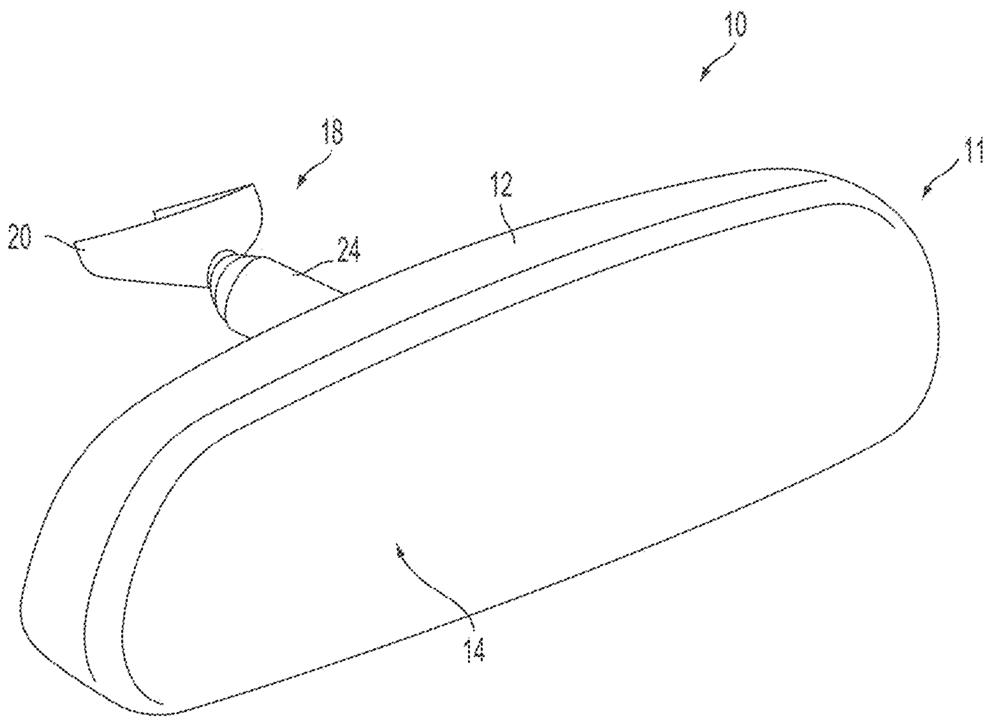
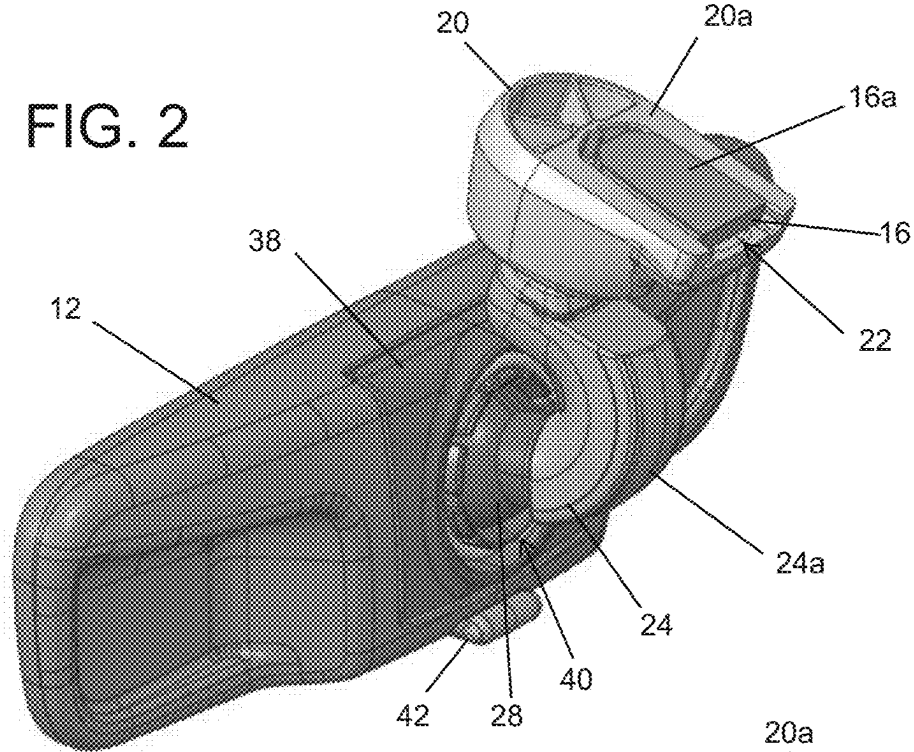


FIG. 1

FIG. 2



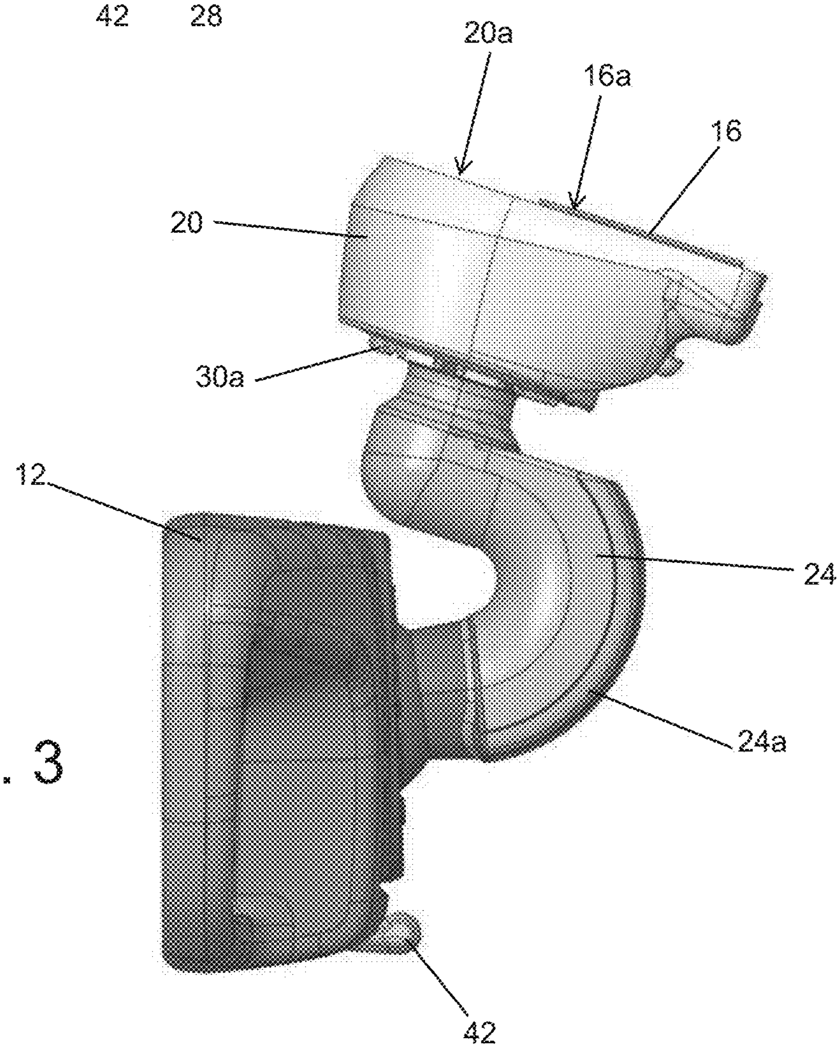
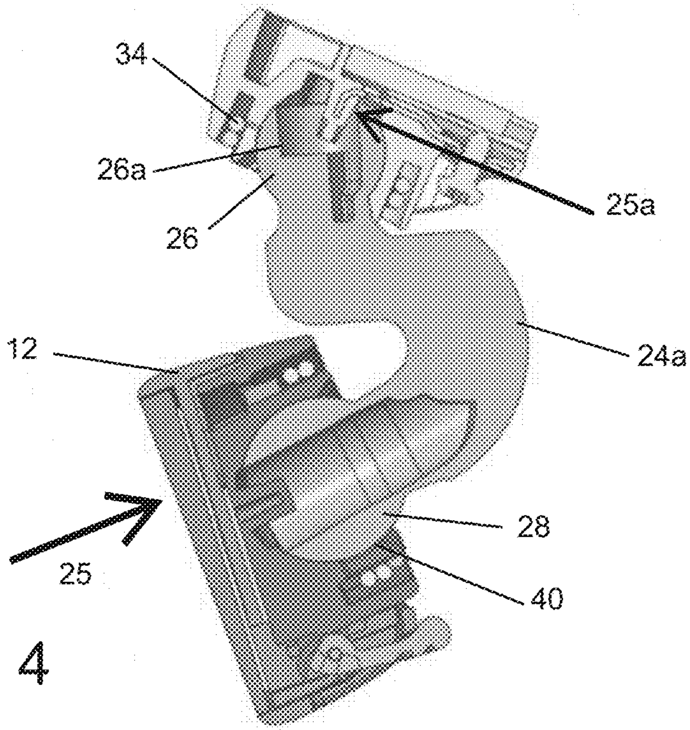


FIG. 3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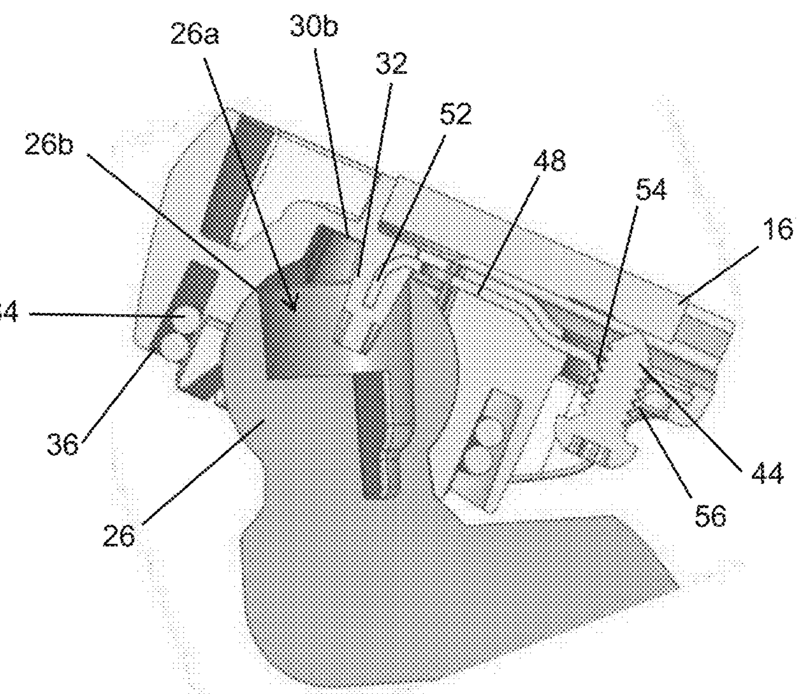
24a
12
28
25
40

FIG. 4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56
26

FIG. 5

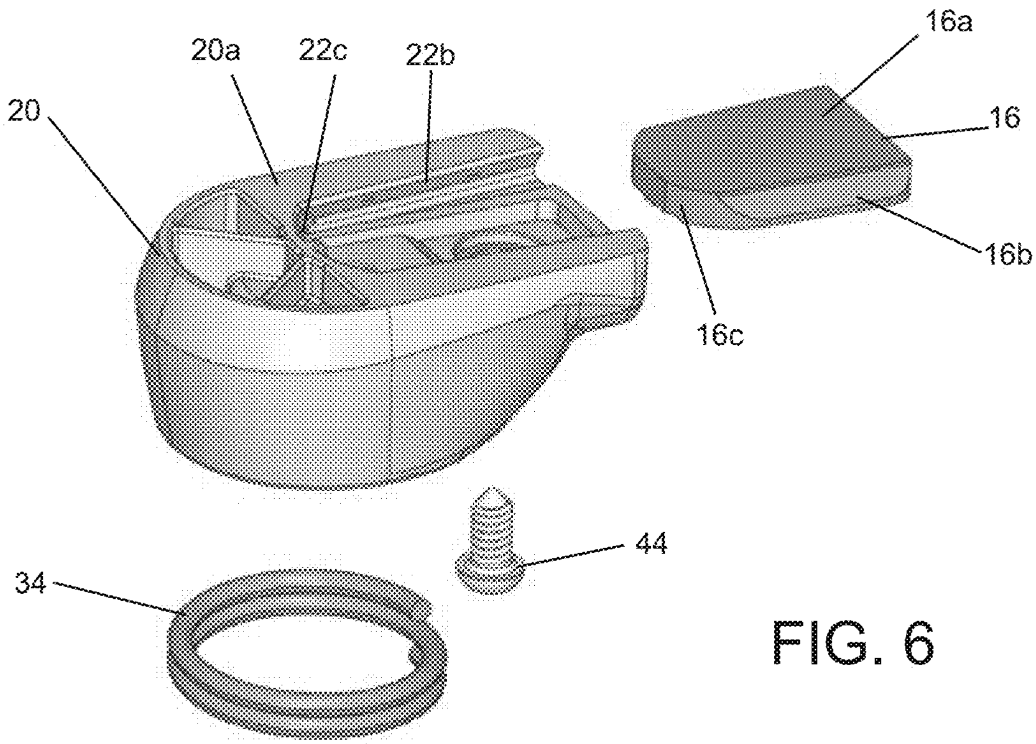


FIG. 6

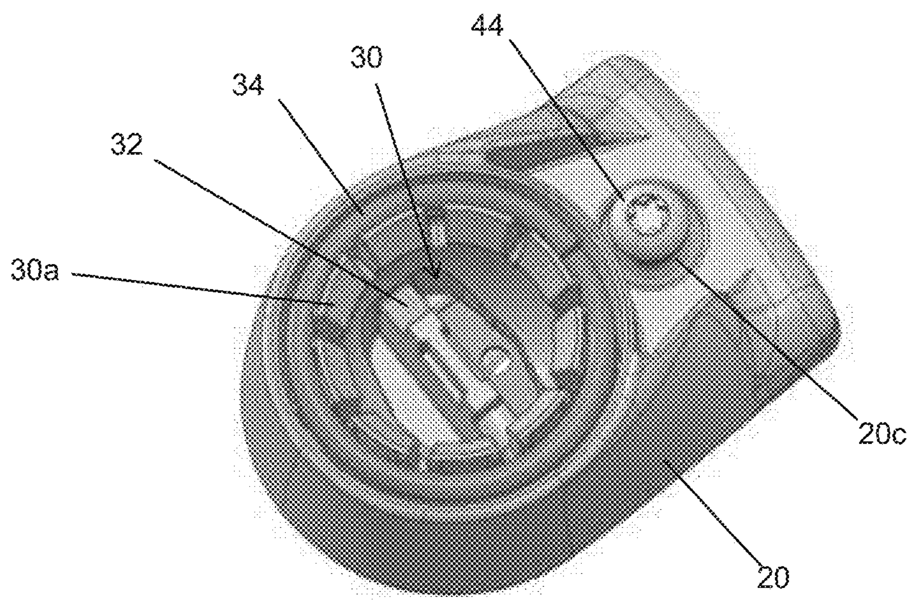


FIG. 7

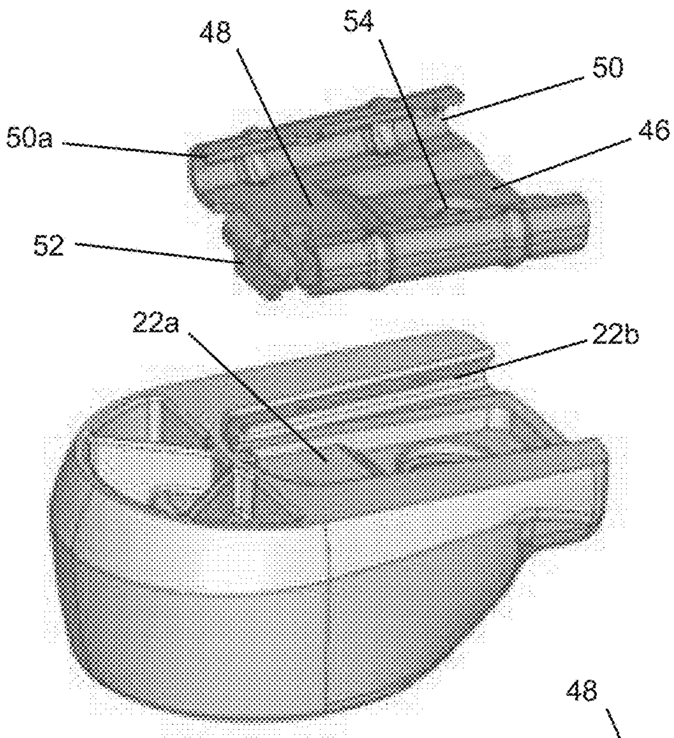


FIG. 8

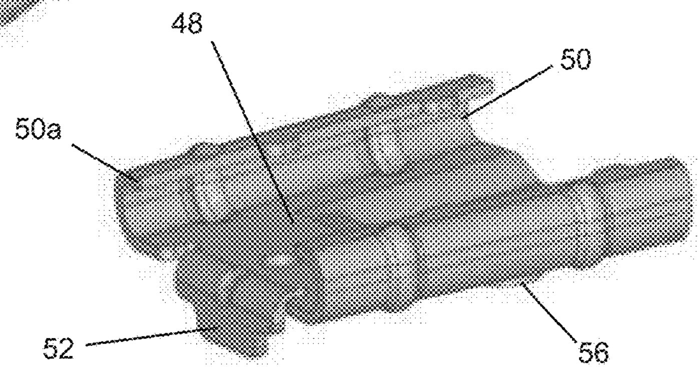


FIG. 9

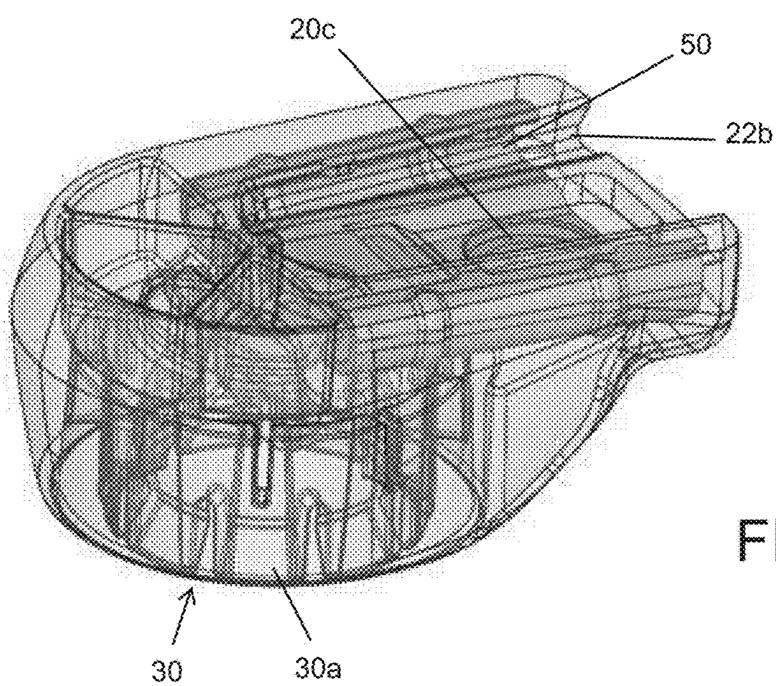


FIG. 10

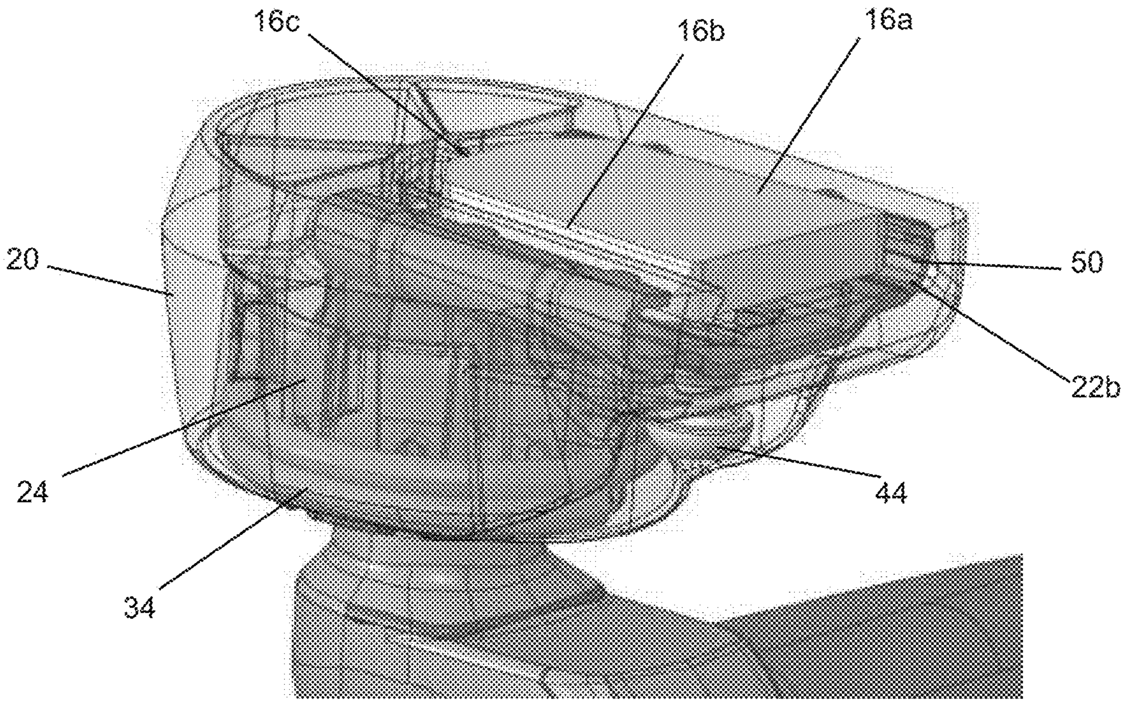
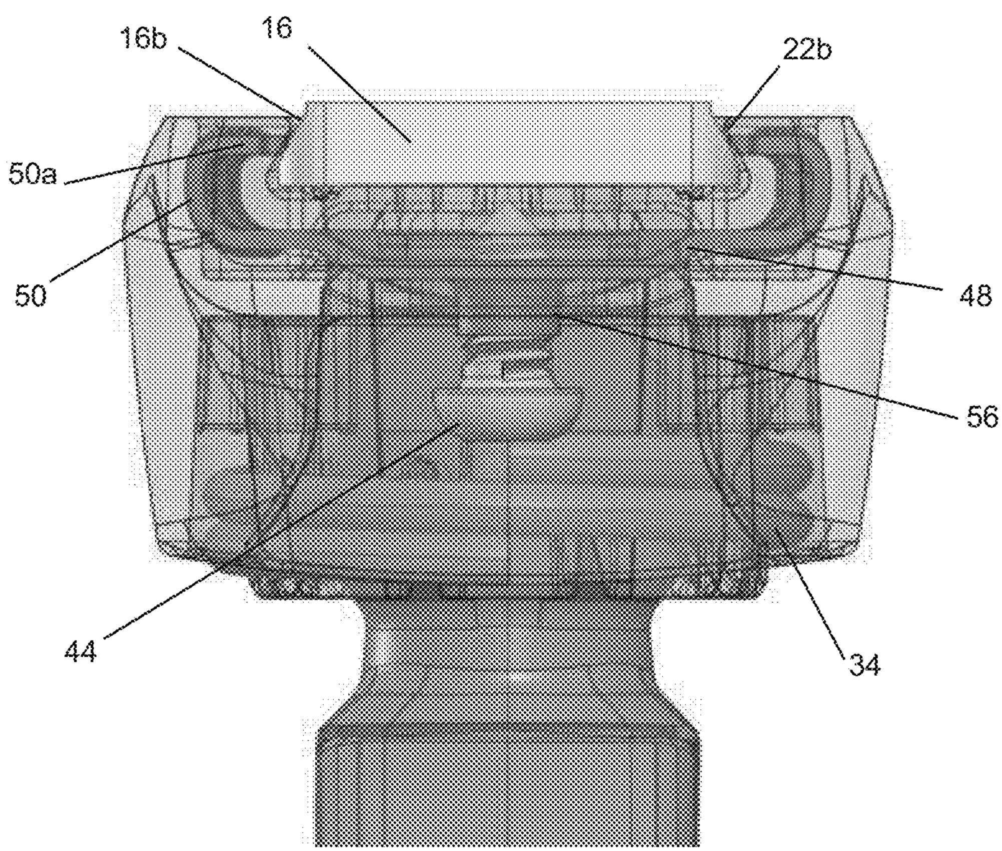


FIG. 11



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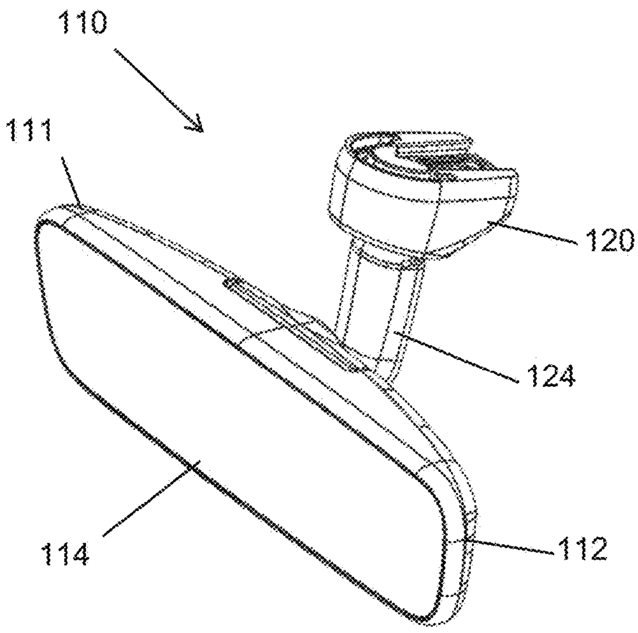


FIG. 13

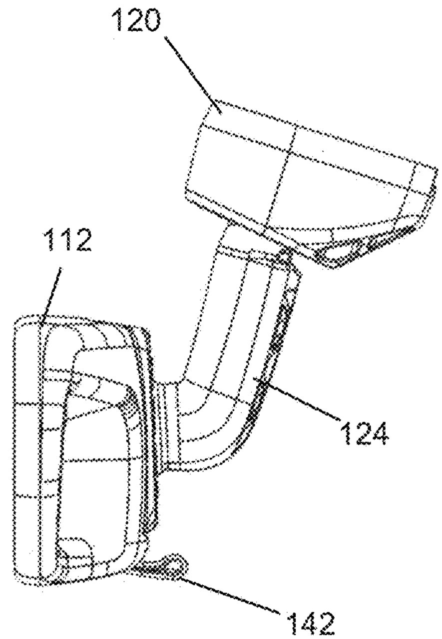


FIG. 14

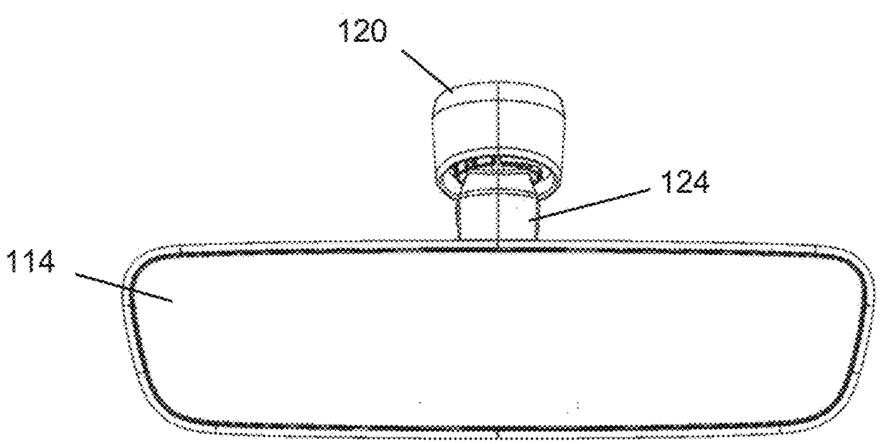
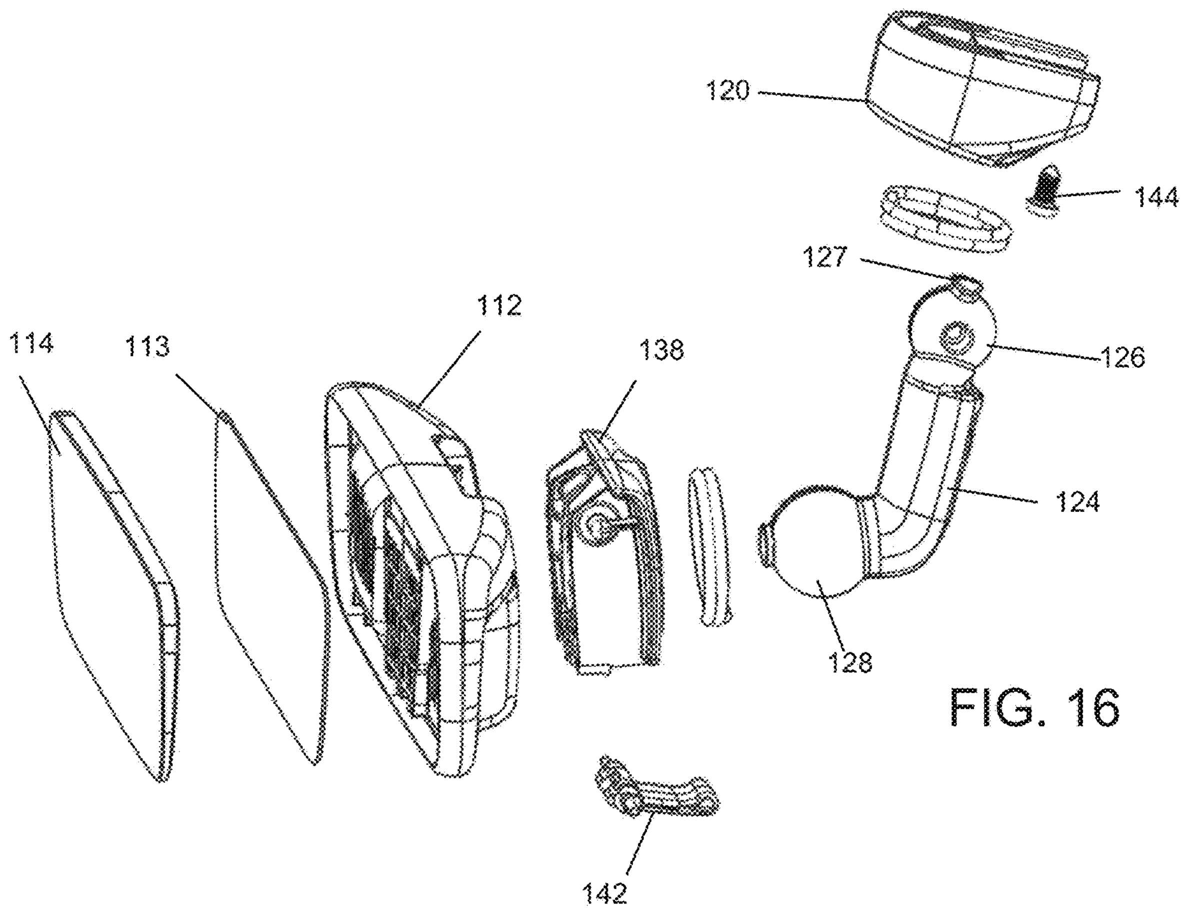


FIG. 15



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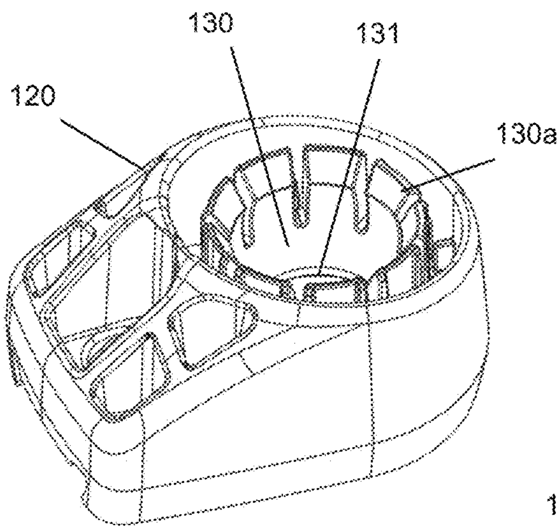

FIG. 17

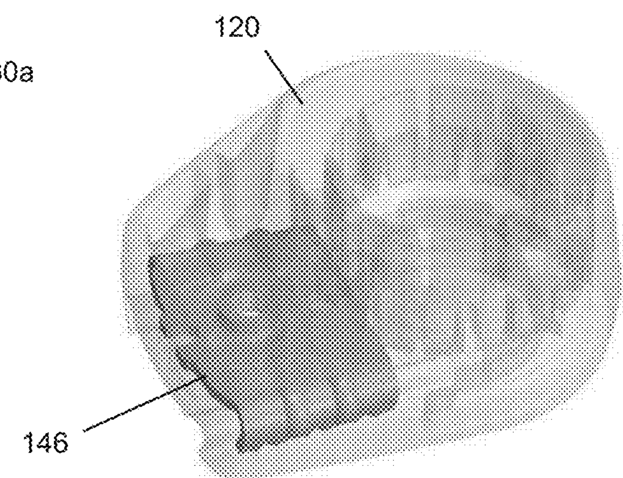

FIG. 18

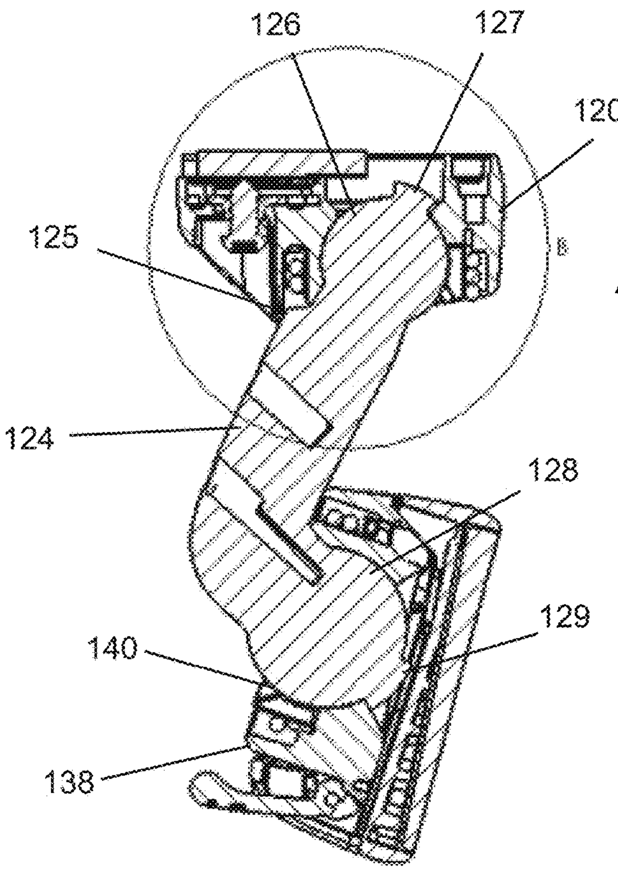

FIG. 19A

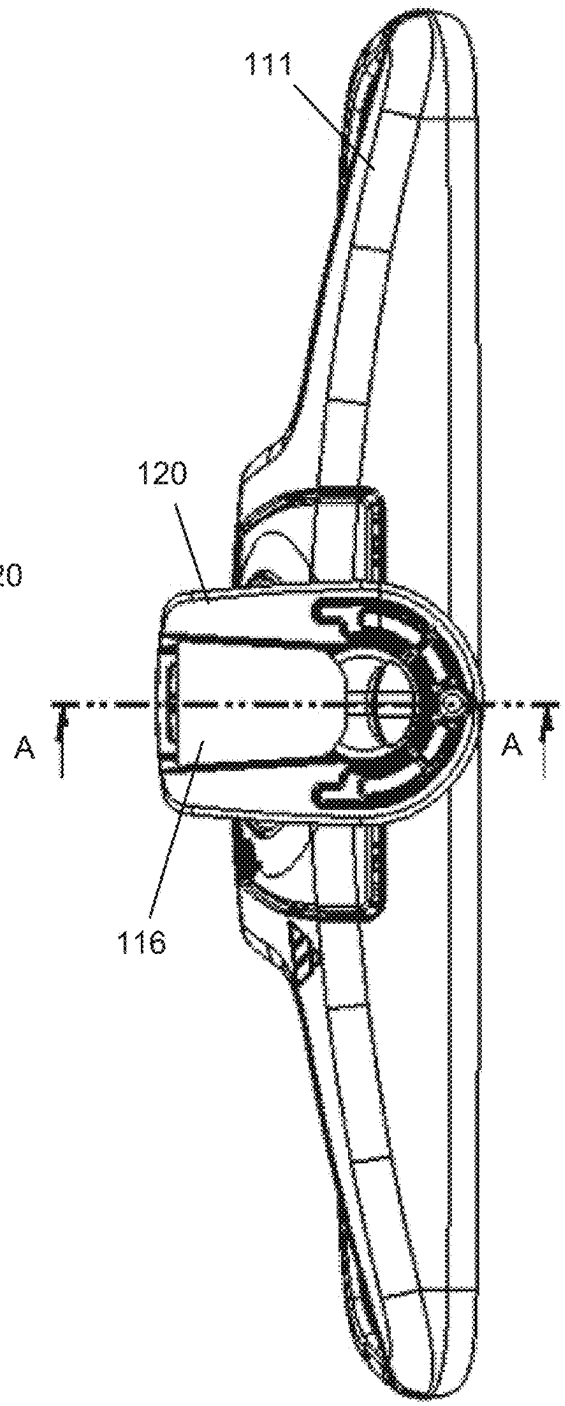

FIG. 19

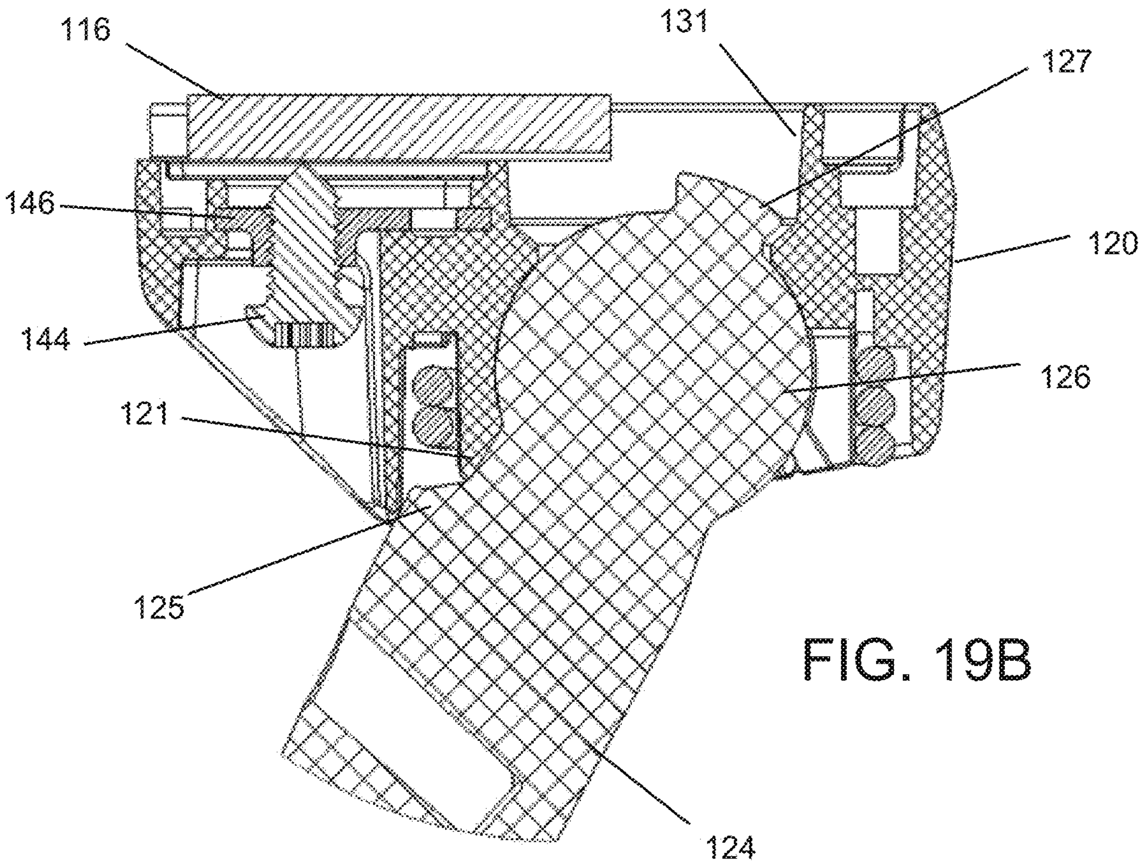

FIG. 19B

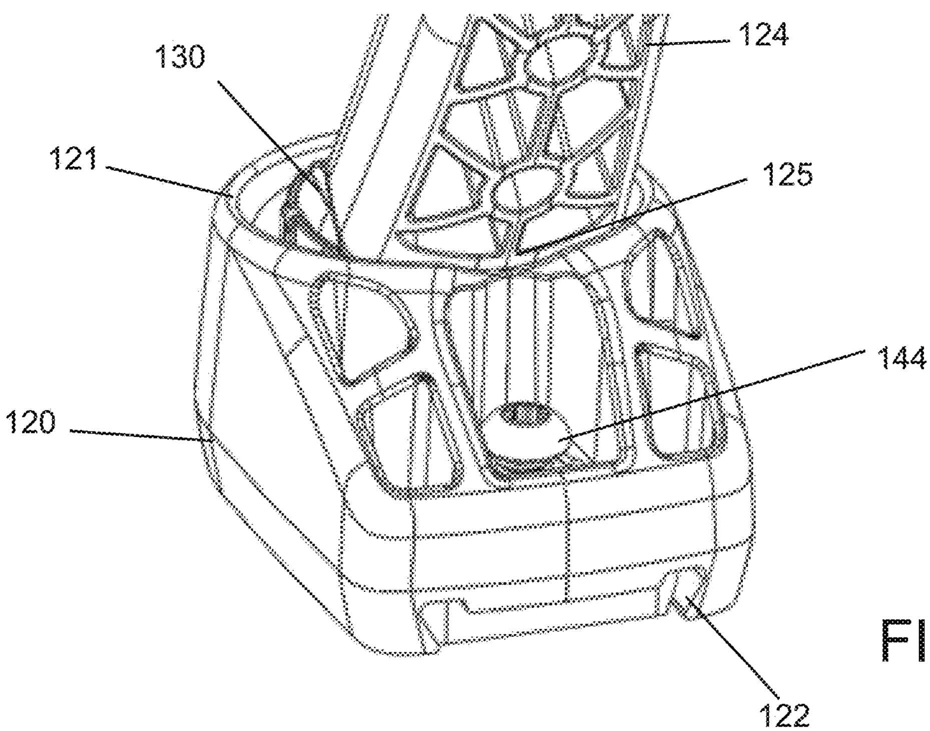

FIG. 20

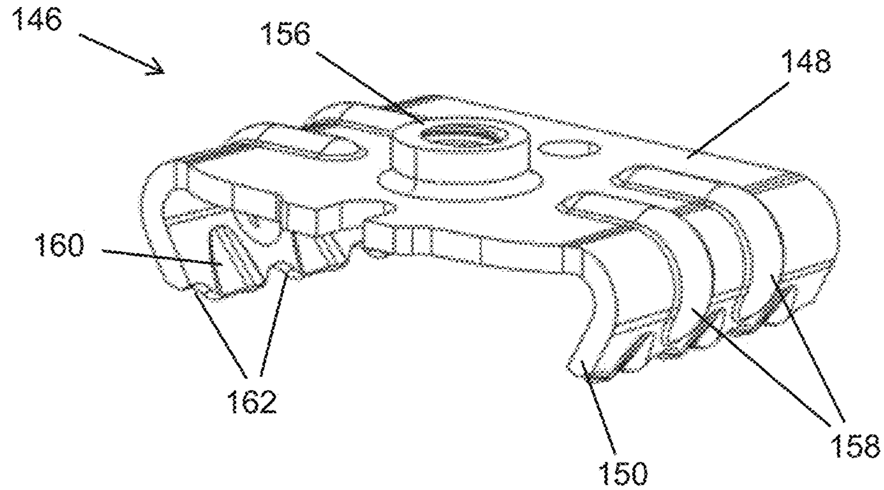

FIG. 21

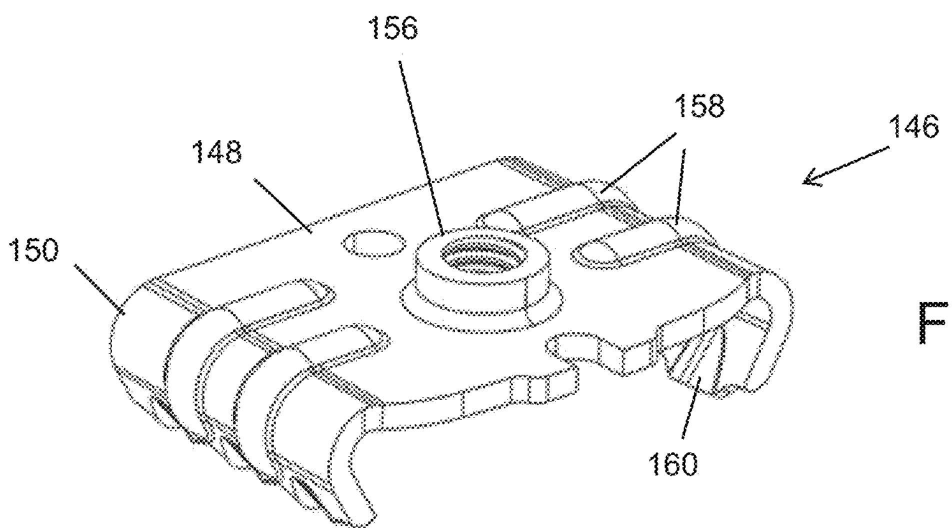

FIG. 22

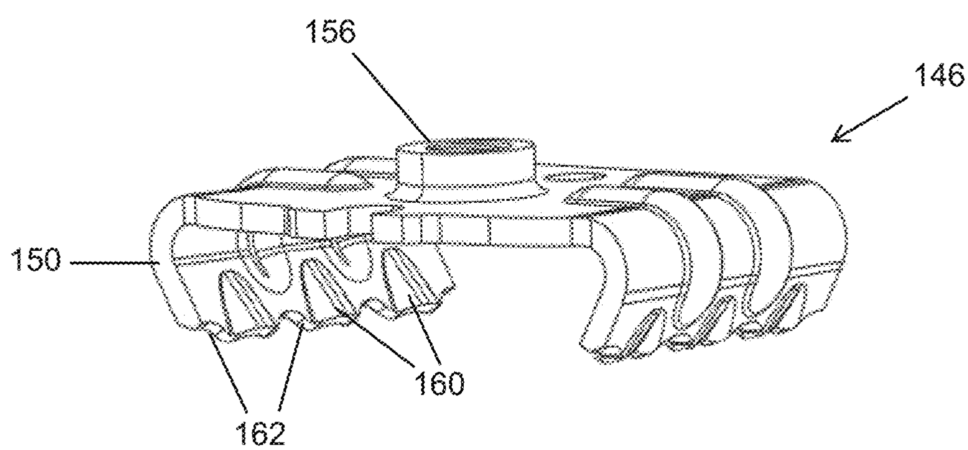

FIG. 23

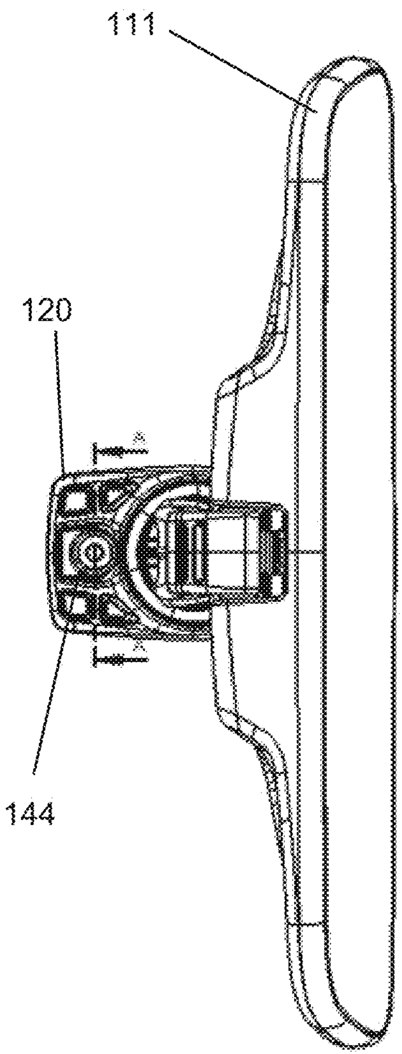

FIG. 24

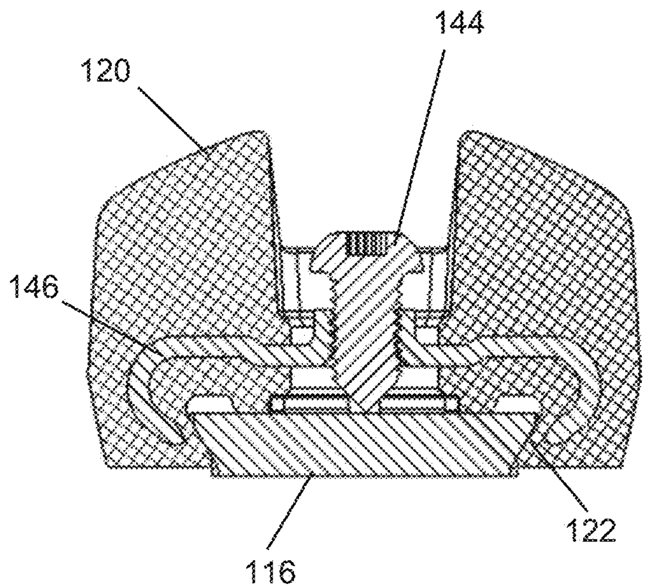

FIG. 24A

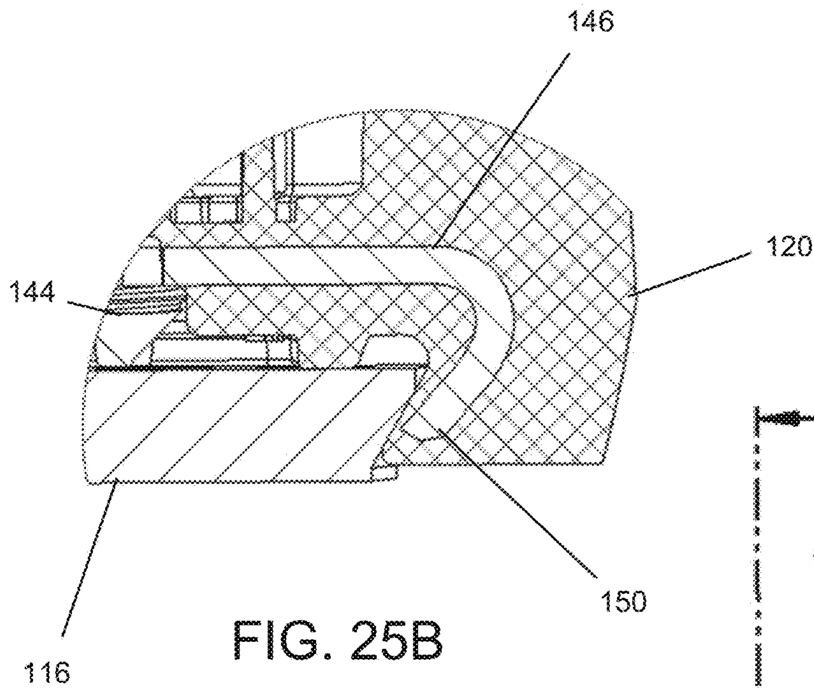

FIG. 25B

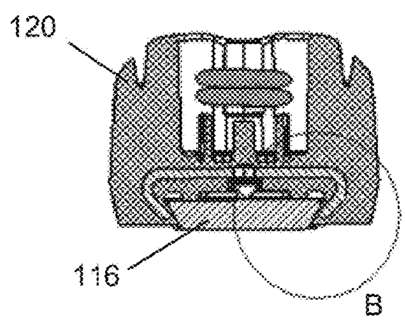

FIG. 25A

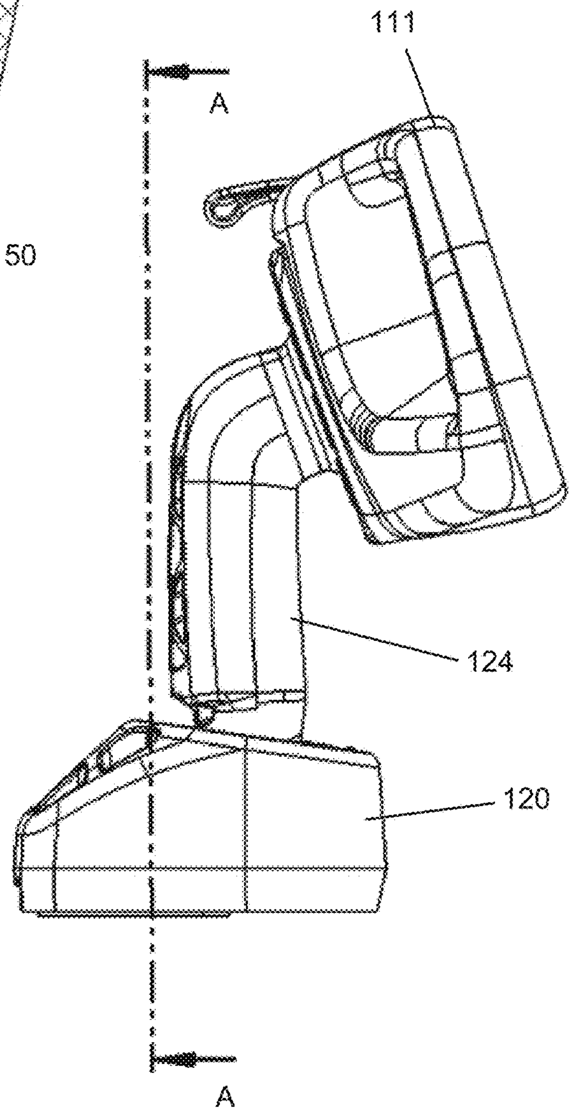

FIG. 25

INTERIOR REARVIEW MIRROR ASSEMBLY WITH OVERMOLDED METAL MOUNT AND ANTI-CAMOUT TAB

CROSS REFERENCE TO RELATED APPLICATION

The present application claims the filing benefits of U.S. provisional application Ser. No. 63/201,236, filed Apr. 20, 2021, which is hereby incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

The present invention relates generally to the field of interior rearview mirror assemblies for vehicles.

BACKGROUND OF THE INVENTION

It is known to provide a mirror assembly that is adjustably mounted to an interior portion of a vehicle, such as via a double ball pivot or joint mounting configuration where the mirror casing and reflective element are adjusted relative to the interior portion of a vehicle by pivotal movement about the double ball pivot configuration. The mirror casing and reflective element are pivotable about either or both of the ball pivot joints by a user that is adjusting a rearward field of view of the reflective element.

SUMMARY OF THE INVENTION

An interior rearview mirror assembly is configured to mount at an interior portion of a vehicle via a mounting arm attached at a mirror head of the mirror assembly and pivotally attached at a socket base or mirror mount. The mounting arm is pivotally attached at the mirror mount via a ball member of the mounting arm received within a socket of the mirror mount. An anti-camout tab protrudes from an interior surface of the socket and is at least partially received within a recess of the ball member. The anti-camout tab prevents the ball member from rotating within the socket beyond a point where the ball member engages the anti-camout tab. A metallic support structure at the mirror mount provides at least a portion of the anti-camout tab to provide increased strength and durability to the plastic material of the mirror mount.

A vehicular interior rearview mirror assembly includes a mounting assembly including a mirror mount and a mounting arm. The mirror mount is configured to attach at an interior portion of a vehicle equipped with the vehicular interior rearview mirror assembly. A mirror head includes a mirror reflective element. The mirror head is pivotally mounted at the mounting arm and the mounting arm is pivotally attached at the mirror mount. The mounting arm is pivotally attached at the mirror mount via a pivot joint that includes a ball member of the mounting arm received in a socket element of the mirror mount. The mirror mount includes an anti-camout element protruding from an interior surface of the socket element and that is at least partially received in a recess of the ball member to limit rotation of the ball member within the socket element beyond where the ball member engages the anti-camout element. The mirror mount includes a metallic support element forming at least a portion of the anti-camout element.

These and other objects, advantages, purposes and features of the present invention will become apparent upon review of the following specification in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view of an interior rearview mirror assembly;

FIG. **2** is a perspective view of an interior rearview mirror assembly with a double pivot joint mounting construction configured to attach at a mirror button for mounting at an interior portion of a vehicle;

FIG. **3** is a side view of the interior rearview mirror assembly of FIG. **2**;

FIG. **4** is a cross-sectional view of the interior rearview mirror assembly of FIG. **2**;

FIG. **5** is a cross-sectional view of the mounting arm and mirror mount of FIG. **2**;

FIG. **6** is an exploded view of the mirror mount and mirror button;

FIG. **7** is a perspective view of the mirror mount of FIG. **6**;

FIG. **8** is a perspective view of a mirror mount and a metallic support structure;

FIG. **9** is a perspective view of the metallic support structure of FIG. **8**;

FIG. **10** is a perspective view of the mirror mount overmolded over the metallic support structure;

FIG. **11** is a perspective view of the mirror mount of FIG. **10**, shown with a mirror button received at a button receiving portion of the mirror mount;

FIG. **12** is another view of the mirror mount and mirror button of FIG. **11**;

FIGS. **13-15** are views of another interior rearview mirror assembly with a double pivot joint mounting construction configured to attach at a mirror button for mounting at an interior portion of a vehicle;

FIG. **16** is an exploded view of the interior rearview mirror assembly of FIGS. **13-15**;

FIGS. **17** and **18** are perspective views of the mirror mount of FIGS. **13-15**;

FIG. **19** is a top view of the interior rearview mirror assembly of FIGS. **13-15**;

FIG. **19**A is a cross-sectional view taken along Line A-A of FIG. **19**;

FIG. **19**B is an enlarged view of Portion B of FIG. **19**A;

FIG. **20** is a perspective view of the mounting arm received in the mirror mount and pivoted to a maximum pivot position;

FIGS. **21-23** are perspective views of the metallic support structure of the mirror mount of FIGS. **13-15**;

FIG. **24** is a bottom view of the interior rearview mirror assembly of FIGS. **13-15**;

FIG. **24**A is a cross-sectional view taken along Line A-A of FIG. **24**;

FIG. **25** is a side view of the interior rearview mirror assembly of FIGS. **13-15**;

FIG. **25**A is a cross-sectional view taken along Line A-A of FIG. **25**; and

FIG. **25**B is an enlarged view of Portion B of FIG. **25**A.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

Referring now to the drawings and the illustrative embodiments depicted therein, an interior rearview mirror assembly **10** for a vehicle includes a mirror head **11** adjustably mounted at a mounting structure or assembly **18** (FIG. **1**). The mirror head **11** includes a mirror casing **12** and a reflective element **14** positioned at a front portion of the casing **12**. In the illustrated embodiment, the mirror assembly **10** is configured to be adjustably mounted to an interior portion of a vehicle (such as to an interior or in-cabin surface of a vehicle windshield or a headliner of a vehicle or the like) via the mounting structure or mounting configuration or assembly **18**. The mirror reflective element **14** comprises a prismatic mirror reflective element that is flipped between a daytime position and a nighttime position via a toggle element **42** that controls a toggle mechanism **38**. Optionally, the mirror assembly may have a variable reflectance mirror reflective element that varies its reflectance responsive to electrical current applied to conductive coatings or layers of the reflective element. The mirror mounting assembly **18** includes a socket base or mounting base or mirror mount **20** that is configured to mount at a mounting plate or mirror mounting button **16** attached at an interior portion of the vehicle, such as at a mounting button adhered at the in-cabin side of the vehicle windshield. The mirror mount **20** receives a mounting arm **24** that is attached to the rear of the mirror head **11**. The mounting arm **24** includes a ball member **26** pivotally received within a socket **30** of the mirror mount **20** and the mirror mount **20** includes an anti-camout tab or anti-camout element **32** protruding from an inner surface of the socket **30** and within a recess of the ball member **26** to limit pivotal movement of the ball member **26** within the socket **30** to prevent the ball member **26** and mounting arm **24** from pulling out or detaching or releasing from the socket and mirror mount. As described further below, the mirror mount includes a metallic support structure **46** integrally molded with the mirror mount **20** and forming at least a portion of the anti-camout tab **32** to provide added strength and durability to the mirror mount **20**.

When installed at an in-cabin side of the windshield, the mirror mounting button **16** may be adhesively attached to the windshield and the mirror mount **20** engages the mirror button **16**. For example, the mirror mount **20** may snap attach to the mirror button **16** or slidably engage the mirror button **16**. In the illustrated embodiment (and with reference to FIGS. **2** and **3**), the socket base or mirror mount **20** receives the wedge-shaped mirror mounting button **16** at a button receiving portion **22** of the mirror mount **20**. A locking element or set screw **44** engages the mirror button **16** through the mirror mount **20** to retain the mirror mount **20** at the mirror mounting button **16**. The mirror mount **20** receives the first ball member **26** of the mounting arm **24** in the socket **30** of the mirror mount **20** and the mounting arm **24** connects to the mirror head **11** via a second ball member **28** received in a socket **40** in the toggle mechanism **38** of the mirror head.

The mounting arm **24** comprises a plastic 2-ball linkage piece that has the first ball member **26** at one end of a longitudinally curved or bent arm portion **24*a*** and the second ball member **28** disposed at the opposite end of the arm portion **24*a***. The first ball member **26** is pivotally received at the socket **30** of the mirror mount **20** (and retained therein via a coil spring **34** and flexible tabs **30*a*** that define the interior surface of the socket **30**). The mounting arm **24** may be fixedly attached at the mirror head **11** or the second ball member **28** at the other end of the arm portion **24*a*** may be pivotally received at the socket **40** of the toggle mechanism **38** of the mirror head **11**. The plastic 2-ball linkage mounting arm **24** may comprise a single or unitary piece of plastic (such as made via injection molding of a plastic material). The double ball-pivot joint of the mounting arm **24** allows a user to pivot the mirror reflective element at the junction between the mirror casing and mounting arm and/or at the junction between the mounting arm and the mirror mount. Thus, the mirror reflective element may be pivotable by a user to a number of different configurations so as to provide a desired rearward view to a driver of the vehicle.

The socket **30** receives the first ball member **26** of the mounting arm **24** to provide a first pivot joint for pivoting the mounting arm **24** and mirror head **11** relative to the mirror mount **20**. The flexible elements or tabs **30*a*** (that, together with the inner surface **30*b*** of the mirror mount, define the socket **30**), during assembly and/or installation of the mounting arm at the mirror mount, flex to allow the ball **26** of the mounting arm **24** to be pressed into the socket **30**. As the ball **26** is received within the socket **30**, the flexible tabs **30*a*** snap back toward their initial state to retain the ball member **26** in the socket **30**. As shown in the illustrated embodiment (and with reference to FIGS. **4**-**7**), the coil spring **34** is inserted into a circular slot **36** surrounding or circumscribing the socket **30** of the mirror mount to provide a constricting or biasing force on an external surface of the flexible tabs **30*a*** to further retain the ball member in the socket. The socket **40** at the toggle mechanism **38** is similarly oriented with flexing tabs and a coil spring for pivotally receiving the second ball member **28** to provide a second pivot joint, and the flexing tabs and/or coil spring at each socket may be selected to provide greater resistance to pivoting at one end of the mounting arm than the other. For example, the mounting arm **24** may have greater resistance to pivoting at the socket **30** of the mirror mount **20** as compared to the resistance to pivoting of the socket **40** of the toggle mechanism and mirror casing at the other end of the mounting arm. The ball members and respective sockets may be formed to resist or limit pivoting within a range of selected or appropriate pivotal movement.

As shown in FIGS. **4** and **5**, the anti-camout element **32** protrudes from an inner surface of the socket **30** so that the anti-camout element **32** is at least partially received within a recess **26*a*** of the first ball member **26** when the first ball member **26** is disposed in the socket **30**. The anti-camout element **32** prevents the ball joint from rotating or pivoting beyond a given point where the mounting arm **24** may inadvertently detach or pull out of the socket **30**. As will be discussed further below, the plastic mirror mount **20** (formed from such material as a 30 percent glass filled Polyoxymethylene (POM) or acetal or the like) includes a metallic support structure or element or tab **52** that forms at least a portion of the anti-camout element **32** and provides a rigid internal element for the mirror mount **20**.

The ball member **26** of the mounting arm **24** includes an interior surface defining the recess **26*a*** in the upper portion of the ball member that is received within the socket **30** of the mirror mount **20**. When received in the socket **30**, the ball member **26** is free to pivot within the socket (to pivot or position the mirror reflective element as desired by the user), to a degree permitted by the anti-camout tab **32**. As the mounting arm **24** is pivoted to position the mirror reflective element at various positions, the interior surface or edge **26*b*** of the ball member recess **26*a*** engages the anti-camout tab **32** to prevent the ball member from pivoting beyond a point where the ball member engages the anti-camout tab. The anti-camout tab **32** is positioned and sized to allow a significant degree of movement of the ball member **26** and therefore of the mounting arm **24** and mirror head and also to limit or preclude the ball member **26** from being rotated or pivoted beyond a point where the ball member could fall or pull out of the socket. For example, if the ball joint is rotated beyond a point so that the edge **26***b* of the ball member recess **26***a* is external the socket (or past the edge of the socket), the ball member could pull out of the socket.

In the illustrated embodiment, the anti-camout tab is centrally aligned with a vertical axis of the socket **30** of the mirror mount to provide an equally distributed range of motion for the ball member **26** within the socket. Optionally, the anti-camout tab and/or recess **26***a* of the ball member may be configured to provide other ranges of motion. For example, the anti-camout tab may be positioned within the socket in such a way as to allow greater range of motion in one direction and a lesser range of motion in the opposite direction. Placing the anti-camout tab further back in the socket (further from the front surface of the mirror reflective element) may allow the mounting arm to pivot further upward (relative to a neutral position) via a ball joint that has a greater corresponding range of motion, but pivot to a lesser degree downward via the ball joints corresponding shorter range of motion.

As the ball member **26** is rotated or pivoted within the socket, the top edge **26***b* of the recess **26***a* contacts the outer surface or edge of the anti-camout tab **32** to prevent the ball member from further rotating or pivoting within the socket beyond the point where the edge **26***b* or inner surface of the recess engages the anti-camout tab **32**. If a user attempts to pivot the mirror reflective element beyond the point permitted by the anti-camout tab **32**, a force may be exerted through the ball member **26** onto the anti-camout tab **32**. For example, see FIG. **4** which depicts a force **25** (represented by an arrow) applied at the mirror reflective element (such as to reposition the mirror reflective element to provide a desired view) and the resulting force **25***a* applied via the mounting arm **24** and ball member **26** upon the anti-camout tab **32**. Over time, repeated forces acting upon the anti-camout tab **32** may bend or degrade or otherwise damage the plastic portion of the tab. Thus, the metallic support structure **46** that forms at least a portion of the anti-camout tab **32** provides additional rigid support to increase the lifespan of the tab.

The metallic support structure **46** is overmolded by the unitary plastic element of the mirror mount **20** and provides rigid support to the various portions of the mirror mount, including the anti-camout tab **32**. The tab **52** of the metallic support structure **46** extends along and is overmolded by the plastic anti-camout tab **32** to provide a substantially rigid anti-camout tab that will not bend or break when a force is applied to the mounting arm at a point where the anti-camout tab **32** contacts the ball member to limit or preclude pivoting of the mounting arm past a given orientation relative to the mounting base.

As shown in FIGS. **8-12**, the metallic support structure **46** comprises a piece of stamped steel having a substantially flat base portion **48** with curved arms or rails or lateral receiving channels **50** extending along opposite edges of the base portion. The tab **52** extends from one end of the base portion, and the pair of receiving channels **50** are configured to provide support to side walls of the button receiving portion **22** of the mirror mount **20** that receives the sides of the mirror mounting button when the mirror mount is attached at the mounting button at the interior portion of the vehicle (e.g., at the in-cabin side of the vehicle windshield). A threaded through hole **54** in the base portion **48** aligns with a through hole or passageway **20***c* in the body of the mirror mount **20** and threadably receives the set screw **44** therein (such as within a screw receiving protrusion **56** comprising a threaded inner surface) to retain the support structure **46** at the mirror mounting button.

As discussed above, the support tab **52** protrudes from an end region of the base portion **48** of the support structure **46** and forms at least a portion of the anti-camout tab **32** to provide structural support to the anti-camout tab and resist forces placed on the anti-camout tab from the ball member. Optionally, the support tab **52** may protrude from a lower surface of the support structure **46**. The support tab **52** protrudes from the base portion **48** and is substantially perpendicular to the base portion of the support structure. In the illustrated embodiment (such as seen in FIG. **5**), the support tab **52** of the support structure **46** is contained or overmolded entirely within the body of the anti-camout tab **32** and provides a rigid core for the anti-camout tab to prevent the anti-camout tab **32** from bending or breaking when excessive or repeated force is applied via pivoting of the ball member **26**. Although shown as overmolded with the plastic material of the mirror mount such that the entirety of the support tab is within the body of the anti-camout tab, the support tab of the support structure may provide any portion or the entirety of the anti-camout tab. For example, the support tab **52** is shown as being centrally located within the anti-camout tab **32** (to provide a rigid core of the anti-camout tab), but the support tab may be positioned within the overmolded plastic material of the mirror mount to provide one or more of the outer surfaces of the anti-camout tab. Optionally, the support tab **52** may comprise the entirety of the anti-camout tab **32** such that there is no plastic overmolded onto the metallic support tab. Thus, the mirror assembly includes a tab on the steel support that reinforces the plastic anti-camout feature located in the socket and/or comprises the anti-camout tab of the mirror mount. The steel tab allows higher forces to be applied to the mirror and/or ball joint in its maximum adjustment travel before damaging the plastic feature and prevents the ball from being forced out of the socket.

The metallic support structure **46** may be designed to also support stresses in the dove tail interface between the mirror mount **20** and the mirror button **16**. As shown in FIGS. **11** and **12**, the mirror mount **20** slidably engages the mirror button **16** at the button receiving portion **22** of the mirror mount. The button receiving portion **22** is defined by a recess or slot in the top surface **20***a* of the mirror mount **20** and inwardly angled side surfaces **22***b* along opposing sides of the slot. The inwardly angled side surfaces **22***b* provide receiving rails or guides for correspondingly angled side surfaces **16***b* of the mirror button **16**. The corresponding angled relationship between the side surfaces of the button receiving portion and the mirror button form a dove tail joint so that the weight of the mirror mount (and attached mirror reflective element) is supported vertically by the mirror button. The curved arms or wings or lateral support tabs or receiving channels **50** of the metallic support structure **46** are integrally molded within the mirror mount **20** to align with and provide support at the dove tail joint.

The lateral support tabs **50** protrude upward from the base portion **48** within the body of the mirror mount **20** along the side surfaces **22***b* of the button receiving portion **22** and comprise inward facing curved or folded portions **50***a* that at least partially wrap around the angled side surfaces **22***b*. The support structure **46** is positioned within the body of the mirror mount **20** such that the top surface **22***a* of the button receiving portion **22** lies on a plane above the base portion **48** and below the top of the curved portion **50***a* so that the force from supporting the weight of the mirror mount and mirror assembly (and any other forces that may be felt at the mirror mount and/or mirror button) may be dispersed within the support structure, increasing the stability of the dove tail joint. Additionally, the metallic support structure helps resist dimensional changes and breaking or cracking of the mirror mount which can be caused by excessive force and/or thermal cycling. Thus, the metallic support structure will increase the dimensional stability of the dove tail joint through environmental and thermal changes which will reduce the risk of looseness that could occur from plastic relaxation and thermal expansion. Furthermore, the metallic support structure reduces the risk of detachment of the mirror mount from the windshield button when exposed to excessive forces (such as during ECE pendulum impact testing).

Once the mirror mount **20** slidably engages the mirror button **16** (such as to a point where the front rounded edge **16***c* of the mirror button aligns with or is stopped by the correspondingly rounded front edge **22***c* of the button receiving portion **22**), the locking element or set screw **44** may be threaded into and through the mirror mount to retain the mirror mount at the mirror button and prevent the mirror mount from slidably detaching from the mirror button. As shown in the illustrated embodiment (such as FIGS. **11** and **12**), the set screw **44** threadably engages the mirror mount and/or metallic support structure via the through hole **20***c* established in an outer lower surface of the mirror mount **20** and the threaded hole **54** through the metallic support structure. The set screw **44**, when threaded into the threaded hole of the metallic support structure, may engage the mirror button **16** and provide a clamping force to press the mirror button into the side surfaces **22***b* of the button receiving portion.

The through hole **54** of the support structure **46** is formed through a screw engaging protrusion **56** at the base portion **48** that aligns with the through hole **20***c* established in the mirror mount. The support structure may be overmolded with the mirror mount in such a way that the internal surface of the support structure through hole and/or screw engaging protrusion is exposed to be threadably engaged with the set screw. Thus, the set screw may thread into the steel which is harder and more wear resistant than the glass filled POM material of the mirror mount. By reducing the stress in the threads, the steel support allows for increased set screw torque, which increases clamp load. That is, the steel support allows for higher torque to be applied to the fixing set screw, thereby increasing the clamp load between the mirror mount and the mirror button at the vehicle. This improves vibration performance and reduces the risk of noise caused by loose components, such as during buzz, squeak and rattle (BSR) testing. Additionally, threading into the steel allows for the majority of the clamp stress to be transferred through the steel support structure, further improving the structural integrity and stability of the connection between the mirror mount and mirror mounting button. Steel threads also allow for better durability during repeated installations as compared to threading the set screw into the plastic material of the mirror mount.

Although shown as a set screw, the locking mechanism may be any suitable element for engaging the mirror mount to retain the mirror mount at the mirror button. For example, the locking mechanism may be a clip or latch or biasing member at the button receiving portion that releasably engages the mirror button (such as to provide a quick release function of the mirror mount). Additionally, the steel support may have suitable elements to allow the base to detach from the windshield. For example, the support structure may comprise an outer surface or otherwise be an exterior component that provides an integrated locking mechanism (such as a biasing member), similar to elements discussed in U.S. Pat. No. 10,744,944, which is hereby incorporated herein by reference in its entirety.

Optionally, the mounting arm may comprise one or more anti-camout features so that, when the ball member of the mounting arm is received at the socket of the mirror mount, structure of the mounting arm and/or ball member may engage corresponding structure at the mirror mount and/or socket to preclude the mounting arm and ball member from pivoting beyond a point where the ball member may pull out of the socket. Thus, and as shown in FIGS. **13-25**B, a vehicular interior rearview mirror assembly **110** may include a mirror head **111** (including a mirror reflective element **114** and mirror casing **112** where the mirror reflective element **114** may be adhesively attached at the mirror casing **112** via an adhesive element **113**) that mounts at an interior portion of a vehicle via a mounting assembly **118**. The mirror reflective element **114** comprises a prismatic mirror reflective element that is flipped between a daytime position and a nighttime position via a toggle element **142** that controls a toggle mechanism **138** that snap attaches at the mirror head **111**. The mounting assembly **118** includes a mounting base or mirror mount **120** that attaches at an interior portion of the vehicle, such as at an in-cabin surface of a windshield of the vehicle, and a mounting arm **124**. The mounting arm **124** includes a first ball member **126** that is pivotally received in a socket **130** of the mirror mount **120** and a second ball member **128** pivotally received at the mirror head **111** (such as at a socket **140** of the toggle mechanism **138**).

As shown in FIGS. **16-20**, the mirror mount **120** includes a socket **130** defined by inner surfaces of flexible tabs **130***a*. A recess **131** is formed at an upper inner surface of the socket **130**. The first ball member **126** of the mounting arm **124** includes an anti-camout tab **127** protruding from an upper portion of the ball member **126** so that, when the first ball member **126** is received in the socket **130** of the mirror mount **120**, the anti-camout tab **127** is received in the recess **131**. The anti-camout tab **127** and recess **131** are configured so that, when the mounting arm **124** and mirror head **111** are pivoted to a maximum pivot position (FIGS. **19**A, **19**B, and **20**), the anti-camout tab **127** of the ball member **126** may engage a surface of the mirror mount **120** defining the recess **131**. In other words, the inner surface of the mirror mount **120** defining the recess **131** may define the maximum pivot positions of the mounting arm **124** and mirror head **111**.

Optionally, and such as shown in FIGS. **19**B and **20**, the mounting arm **124** may be configured to engage corresponding structure of the mirror mount **120** at a maximum pivot position that is less than the maximum pivot position defined by the anti-camout tab **127** and recess **131**. In other words, when the mirror head **111** and the mounting arm **124** are pivoted towards the maximum pivot position, the mounting arm **124** may engage a portion of the mirror mount **120** before the anti-camout tab **127** engages the inner surface of the recess **131**. For example, the mounting arm **124** may include an extension or shelf **125** at the end of the mounting arm **124** nearest the first ball member **126** and the shelf **125** may be configured to engage a lip **121** of the mirror mount **120** surrounding the socket **130** and raised relative to the outer edge of the socket **130**. The anti-camout tab **127** may then engage the inner surface of the recess **131** if the load applied at the lip **121** deforms or bends the lip **121** and the mounting arm **124** pivots further past the maximum pivot position.

That is, when load is applied to the mirror assembly and the mounting arm **124** reaches the end of travel (i.e., the maximum pivot position), stress is transferred to the supporting surface of the lip **121** of the mirror mount **120**. If the load applied increases beyond a certain threshold to cause enough elastic deformation, the anti-camout feature **127** on the ball member **126** makes contact with the mirror mount **120** within the recess **131**. At this point, the stress is distributed between the lip **121** and the recess **131** to optimize the load carrying capacity in each location such that material failure is avoided.

Optionally, the second ball member **128** of the mounting arm **124** that is received at a socket **140** of the mirror head **111** or toggle mechanism **138** may similarly include an anti-camout tab **129** protruding from an outer surface of the ball member **128**. The anti-camout tab **129** of the second ball member **128** may engage corresponding structure of the mirror head **111** to preclude pivotal movement of the ball member **128** beyond where the anti-camout tab **129** engages the structure at the socket **140** of the mirror head **111**.

As shown in FIGS. **21-25**B, the mirror mount **120** includes a metallic support structure **146** within a body of the mirror mount **120** such that the metallic support structure may be overmolded by the unitary plastic element of the mirror mount **120**. The metallic support structure **146** provides rigid support to the button receiving portion **122** of the mirror mount (that receives the mirror button **116** adhesively attached at the windshield of the vehicle to mount the mirror mount **120** at the mirror button **116**). As shown in FIGS. **21-23**, the metallic support structure includes a substantially flat base portion **148** and curved arms or rails **50** extending along opposite edges of the base portion **148**. The curved arms **50** are configured to extend along opposing angled side walls of the button receiving portion **122** to provide support to the dove tail interface between the mirror mount **120** and the mounting button **116**.

The metallic support structure **146** is also configured to receive a locking mechanism, such as a threaded set screw **144**, for retaining the mirror mount **120** at the mounting button **116**. The metallic support structure **146** includes a screw engaging protrusion **156** extending from the base portion **148** and including a threaded through hole that threadably receives the set screw **144**. The screw engaging protrusion **156** may be 43598732.1 exposed exterior the body of the mirror mount **120** for receiving the set screw **116** while the remaining portion of the metallic support structure **146** is contained within the body of the mirror mount **120**. Thus, the set screw **116** directly engages the metallic support structure **146**. Optionally, the set screw **116** may threadably engage the metallic support structure **146** and pass through the metallic support structure **146** to engage the mirror button **116** at the button receiving portion **122** of the mirror mount **120**. That is, the set screw **144** threads directly into the metallic support structure **146**, distributing stress caused by tightening the set screw into the metallic support structure **146**. Threading into the metallic support structure **146** improves thread life for serviceability.

When the locking mechanism **144** engages the metallic support structure **146**, the clamp load applied between the locking mechanism and mounting button **116** is dispersed through the metallic support structure **146** instead of the mirror mount **120** to limit or preclude stress and deformation of the plastic mirror mount **120** and more securely retain the mirror mount at the mounting button **116**. The metallic support structure includes structural supports or embossments **158** at an outer surface of the support structure **146** and spanning at least a portion of the base portion **148** and support rails **150** to preclude outward flexing of the support rails **150**. The embossments **158** wrap around the bend in the metal to improve stiffness of the metallic support structure **146**. The extra stiffness allows higher torque to be applied to the set screw **144** without damaging the mount **120** or mount support structure **146** and reducing the elastic deformation of the mounting interface.

Additionally, dimples or divots **160** may be formed at an inner surface of the metallic support structure **146** along the respective edges of the support rails **150**. The dimples **160** may allow for plastic to flow more easily around the metallic support structure **146** during the manufacturing process, allowing for a thinner layer of plastic between the metallic support structure **146** and the mirror button **116**. Furthermore, recesses or indentations **162** may be formed along the respective edges of the support rails **150** (such as between each of the dimples **160**) to provide a non-linear or wavy edge of the support rails **150**.

In other words, the edge of the metal support structure **146** is formed to improve clearance to the injection molding tool steel and allowing plastic to flow more easily. The features on the metallic support structure **146** are designed to reduce the thickness of 43598732.1 plastic between the windshield button **116** and the metallic support structure **146**. This reduces the loosening effects that can result from thermal expansion and contraction. Keeping the plastic thickness between the metallic support structure **146** and the windshield button **116** small also reduces the effects that dimensional creep could have on loosening and vibration performance. The features on the metallic support structure **146** are designed to provide multiple points of contact between the mirror mount **120** and the windshield button **116**, but also allow plastic to easily flow around the metallic support structure **146** during the injection molding process by only locally reducing the plastic thickness. Thus, the metallic support structure **146** provides improved stiffness and durability to the mounting interface between the mirror mount **120** and the mounting button **116**. The stiffness provided by the metallic support structure **146** reduces stress in the sharp corner of the dove tail mounting slot allowing higher screw torques to be applied without damaging the plastic mount. The added stiffness also allows more load to be applied to the mounted assembly without the mirror unintendedly detaching from the windshield during normal use.

The toggle mechanism may snap attach at the mirror casing and thus may enable quick and easy installation of the mounting assembly to the mirror casing. The toggle mechanism controls flipping of the prismatic mirror reflective element between the daytime position and the nighttime position via the toggle element and receives the mounting arm at the socket of the toggle mechanism. The socket receives the second ball member of the mounting arm in a similar manner as to how the socket receives the first ball member. However, it should be understood that the second ball member of the mounting arm may be received in any suitable manner at the mirror casing or the mounting arm may be fixed or otherwise attached at the mirror casing. For example, the mirror assembly may comprise or utilize aspects of other types of casings or the like, such as described in U.S. Pat. Nos. 7,338,177; 7,289,037; 7,249,860; 6,439,755; 4,826,289 and/or 6,501,387, which are all hereby incorporated herein by reference in their entireties. For example, the mirror assembly may utilize aspects of the flush or frameless or bezelless reflective elements described in U.S. Pat. Nos. 7,626,749; 7,360,932; 7,289,037; 7,255,451; 7,274,501 and/or 7,184,190, which are all hereby incorporated herein by reference in their entireties.

The mirror casing may include a bezel portion that circumscribes a perimeter region of the front surface of the reflective element, or the perimeter region of the front surface of the reflective element may be exposed (such as by utilizing aspects of the mirror reflective elements described in U.S. Pat. Nos. 8,508,831 and/or 8,730,553, and/or U.S. Publication Nos. US-2014-0022390; US-2014-0293169 and/or US-2015-0097955, which are hereby incorporated herein by reference in their entireties).

The mirror assembly may comprise any suitable construction, such as, for example, a mirror assembly with the reflective element being nested in the mirror casing and with a bezel portion that circumscribes a perimeter region of the front surface of the reflective element, or with the mirror casing having a curved or beveled outermost exposed perimeter edge around the reflective element and with no overlap onto the front surface of the reflective element (such as by utilizing aspects of the mirror assemblies described in U.S. Pat. Nos. 7,184,190; 7,274,501; 7,255,451; 7,289,037; 7,360,932; 7,626,749; 8,049,640; 8,277,059 and/or 8,529,108, which are hereby incorporated herein by reference in their entireties) or such as a mirror assembly having a rear substrate of an electro-optic or electrochromic reflective element nested in the mirror casing, and with the front substrate having a curved or beveled outermost exposed perimeter edge, or such as a mirror assembly having a prismatic reflective element that is disposed at an outer perimeter edge of the mirror casing and with the prismatic substrate having a curved or beveled outermost exposed perimeter edge, such as described in U.S. Pat. Nos. 9,827,913; 9,174,578; 8,508,831; 8,730,553; 9,598,016 and/or 9,346,403, and/or U.S. Des. Pat. Nos. D633,423; D633,019; D638,761 and/or D647,017, which are hereby incorporated herein by reference in their entireties (and with electrochromic and prismatic mirrors of such construction are commercially available from the assignee of this application under the trade name INFINITY™ mirror).

As discussed above, the mirror assembly may comprise an electro-optic or electrochromic mirror assembly that includes an electro-optic or electrochromic reflective element. The perimeter edges of the reflective element may be encased or encompassed by the perimeter element or portion of the bezel portion to conceal and contain and envelop the perimeter edges of the substrates and the perimeter seal disposed therebetween. The electrochromic mirror element of the electrochromic mirror assembly may utilize the principles disclosed in commonly assigned U.S. Pat. Nos. 7,274,501; 7,255,451; 7,195,381; 7,184,190; 6,690,268; 5,140,455; 5,151,816; 6,178,034; 6,154,306; 6,002,544; 5,567,360; 5,525,264; 5,610,756; 5,406,414; 5,253,109; 5,076,673; 5,073,012; 5,117,346; 5,724,187; 5,668,663; 5,910,854; 5,142,407 and/or 4,712,879, which are hereby incorporated herein by reference in their entireties.

The mirror reflective element may attach at a backplate within or comprising the mirror casing where the backplate may comprise any suitable construction. Optionally, for example, a common or universal backplate may be provided, whereby the appropriate or selected socket element or pivot element (such as a socket element or such as a ball element or the like) is attached to the backplate to provide the desired pivot joint for the particular mirror head in which the backplate is incorporated. Optionally, when molding the backplate, a different insert may be provided to integrally mold a portion of or all of a ball member or the like (such as a portion of a base of a ball member, whereby the ball member may comprise a metallic ball member that is insert molded at the base and at the rear of the backplate during the injection molding process that forms the backplate, such as by utilizing aspects of the mirror assemblies described in U.S. Pat. Nos. 7,855,755; 7,249,860 and 6,329,925 and/or U.S. Pat. Pub. No. US-2006-0061008, which are hereby incorporated herein by reference in their entireties).

The reflective element and mirror casing are adjustable relative to a base portion or mounting assembly to adjust the driver's rearward field of view when the mirror assembly is normally mounted at or in the vehicle. The sockets and ball members of the mounting structure may utilize aspects of the pivot mounting assemblies described in U.S. Pat. Nos. 6,318,870; 6,593,565; 6,690,268; 6,540,193; 4,936,533; 5,820,097; 5,100,095; 7,249,860; 6,877,709; 6,329,925; 7,289,037; 7,249,860 and/or 6,483,438, and/or U.S. Publication No. US-2018-0297526, which are hereby incorporated herein by reference in their entireties.

Changes and modifications in the specifically described embodiments may be carried out without departing from the principles of the present invention, which is intended to be limited only by the scope of the appended claims as interpreted according to the principles of patent law.

The invention claimed is:

1. A vehicular interior rearview mirror assembly, the vehicular interior rearview mirror assembly comprising:

a mounting assembly comprising a mirror mount and a mounting arm, wherein the mirror mount is configured to mount at a mirror mounting button adhesively attached at an in-cabin side of a windshield of a vehicle equipped with the vehicular interior rearview mirror assembly;

wherein the mirror mount slidably engages the mirror mounting button to attach the mirror mount at the in-cabin side of the windshield of the vehicle;

a mirror head comprising a mirror reflective element;

wherein the mounting arm has a first end and a second end distal from the first end;

wherein the first end of the mounting arm is pivotally attached at the mirror mount via a first pivot joint;

wherein the mirror head is attached at the second end of the mounting arm;

wherein the first pivot joint comprises a ball member at the first end of the mounting arm pivotally received in a socket element of the mirror mount;

wherein the mirror mount comprises an anti-camout element protruding from an interior surface of the socket element, and wherein the anti-camout element is at least partially received in a recess of the ball member to limit pivotal movement of the ball member within the socket element when the ball member engages the anti-camout element;

wherein the mirror mount comprises a metallic support element forming at least a portion of the anti-camout element;

wherein the metallic support element is disposed within a plastic body portion of the mirror mount, and wherein the metallic support element comprises a base portion and a support tab protruding from the base portion;

wherein the base portion of the metallic support element is disposed at a receiving portion of the plastic body portion of the mirror mount;

wherein the receiving portion, when the mirror mount is mounted at the mirror mounting button, slidably engages the mirror mounting button; and

wherein the support tab of the metallic support element is disposed within a plastic portion of the anti-camout element, and wherein the support tab of the metallic support element provides a core portion of the anti-camout element.

2. The vehicular interior rearview mirror assembly of claim 1, wherein the metallic support element is overmolded within the plastic body portion of the mirror mount and wherein the support tab is overmolded within the plastic portion of the anti-camout element.

3. The vehicular interior rearview mirror assembly of claim 1, wherein the receiving portion of the plastic body portion of the mirror mount comprises angled side surfaces that are configured to receive correspondingly angled side surfaces of the mirror mounting button, and wherein the metallic support element comprises lateral support tabs protruding from opposite edges of the base portion of the metallic support element and disposed along the respective angled side surfaces of the receiving portion of the plastic body portion of the mirror mount.

4. The vehicular interior rearview mirror assembly of claim 1, further comprising a locking element which, with the mirror mount mounted at the mirror mounting button, engages the mirror mount and the mirror mounting button to retain the mirror mount at the mirror mounting button.

5. The vehicular interior rearview mirror assembly of claim 4, wherein the locking element comprises a set screw that threadably engages a through hole established in the base portion of the metallic support element.

6. The vehicular interior rearview mirror assembly of claim 5, wherein the through hole of the metallic support element is axially aligned with a through hole established in the mirror mount, and wherein the set screw extends through the through hole of the mirror mount and threadably engages the through hole of the metallic support element.

7. The vehicular interior rearview mirror assembly of claim 1, wherein the mirror head is pivotally attached at the second end of the mounting arm via a second pivot joint.

8. The vehicular interior rearview mirror assembly of claim 1, wherein the mirror mount comprises a flexible lip that at least partially circumscribes the socket element to limit movement of the mounting arm and ball member relative to the mirror mount when the mounting arm engages the flexible lip.

9. The vehicular interior rearview mirror assembly of claim 8, wherein, when the mounting arm initially engages the flexible lip as the mounting arm pivots relative to the mirror mount via the first pivot joint, the ball member does not engage the anti-camout element and, when the mounting arm is further pivoted so that the mounting arm is further moved into engagement with the flexible lip, the flexible lip flexes to allow further movement of the mounting arm and ball member relative to the mirror mount until the ball member engages the anti-camout element.

10. The vehicular interior rearview mirror assembly of claim 1, wherein the plastic body portion of the mirror mount is overmolded over the base portion of the metallic support element, and wherein the metallic support element is disposed within the receiving portion of the plastic body portion.

11. The vehicular interior rearview mirror assembly of claim 10, wherein the base portion of the metallic support element, when the mirror mount is mounted at the mirror mounting button, is spaced from the mirror mounting button.

12. A vehicular interior rearview mirror assembly, the vehicular interior rearview mirror assembly comprising:

a mounting assembly comprising a mirror mount and a mounting arm, wherein the mirror mount is configured to mount at a mirror mounting button adhesively attached at an in-cabin side of a windshield of a vehicle equipped with the vehicular interior rearview mirror assembly;

wherein the mirror mount slidably engages the mirror mounting button to attach the mirror mount at the in-cabin side of the windshield of the vehicle;

a mirror head comprising a mirror reflective element;

wherein the mounting arm has a first end and a second end distal from the first end;

wherein the first end of the mounting arm is pivotally attached at the mirror mount via a first pivot joint;

wherein the mirror head is attached at the second end of the mounting arm;

wherein the first pivot joint comprises a ball member at the first end of the mounting arm pivotally received in a socket element of the mirror mount;

wherein the mirror mount comprises a plastic element overmolded over a metallic support element, and wherein the metallic support element comprises a base portion and a support tab protruding from the base portion;

wherein the base portion of the metallic support element is disposed at a receiving portion of the plastic element of the mirror mount;

wherein the receiving portion, when the mirror mount is mounted at the mirror mounting button, slidably engages the mirror mounting button;

an anti-camout element at the first pivot joint that limits pivotal movement of the ball member within the socket element when the anti-camout element is engaged;

wherein the anti-camout element protrudes from an interior surface of the socket element, and wherein the anti-camout element is at least partially received in a recess of the ball member to limit pivotal movement of the ball member within the socket element when the ball member engages the anti-camout element;

wherein the support tab of the metallic support element is disposed within a plastic portion of the anti-camout element, and wherein the support tab of the metallic support element provides a core portion of the anti-camout element;

wherein the mirror mount comprises a flexible lip that at least partially circumscribes the socket element to limit movement of the mounting arm and ball member relative to the mirror mount when the mounting arm engages the flexible lip; and

wherein, when the mounting arm initially engages the flexible lip as the mounting arm pivots relative to the mirror mount via the first pivot joint, the anti-camout element is not engaged to limit pivotal movement of the ball member within the socket element, and when the mounting arm is further pivoted so that the mounting arm is further moved into engagement with the flexible lip, the flexible lip flexes to allow further movement of the mounting arm and ball member relative to the mirror mount until the anti-camout element is engaged.

13. The vehicular interior rearview mirror assembly of claim 12, wherein the plastic element of the mirror mount is overmolded over the base portion of the metallic support element, and wherein the metallic support element is disposed within the receiving portion of the plastic element.

14. The vehicular interior rearview mirror assembly of claim 13, wherein the base portion of the metallic support element, when the mirror mount is mounted at the mirror mounting button, is spaced from the mirror mounting button.

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